

Customer Behavior Analysis

Insights report: who buys, what they buy, and how they pay

3,900 Customers analysed	\$233,081 Total revenue	\$59.76 Avg purchase amount	79.9% Loyal customers	27.0% Subscribers
------------------------------------	-----------------------------------	---------------------------------------	---------------------------------	-----------------------------

Executive Summary

This report analyses the shopping behavior of 3,900 customers across gender, age, loyalty, subscription, purchase frequency, discounts, payment method, shipping type, product category and season. The headline finding is that the business has a very loyal customer base and evenly balanced spending, but it relies heavily on existing customers and does not appear to gain extra spend from discounts or subscriptions.

Key findings

- **Male customers dominate.** Men account for 68.0% of customers and 67.7% of revenue. Women are 32.0% of customers but spend slightly more per purchase (\$60.25 vs \$59.54).
- **Retention is strong, acquisition is weak.** 79.9% of customers are Loyal and 18.0% Returning, while only 2.1% (83 customers) are New.
- **Age 26-65 is the core market.** These four groups generate 76.2% of revenue. The 46-55 group is the largest (\$45,619, 19.6%); the 66-75 group is the smallest (\$20,904, 9.0%).
- **Discounts do not lift spending.** 43.0% of customers used a discount/promo code, yet their average purchase (\$59.28) is lower than customers who paid full price (\$60.13). 57.3% of revenue came from non-discounted purchases.
- **Subscription helps retention, not basket size.** Only 27.0% of customers are subscribed. Spend per purchase is almost identical (\$59.49 vs \$59.87), but subscribers have a higher repeat-buyer rate (91.0% vs 88.4%).
- **Free Shipping wins.** It is the most used option (675 customers) and generates the highest revenue (\$40,777, 17.5%).
- **Product leaders:** Jewelry (Accessories), Blouse and Pants (Clothing), Sandals (Footwear) and Jacket (Outerwear) lead their categories by orders; Blouse, Shirt and Dress lead by revenue.

Bottom line

Growth will most likely come from **acquiring new customers, growing the female segment and making subscriptions and discounts more profitable**, rather than from changing payment or shipping options, which perform very evenly.

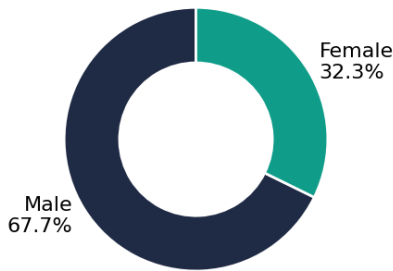
1. Customer Demographics

1.1 Gender

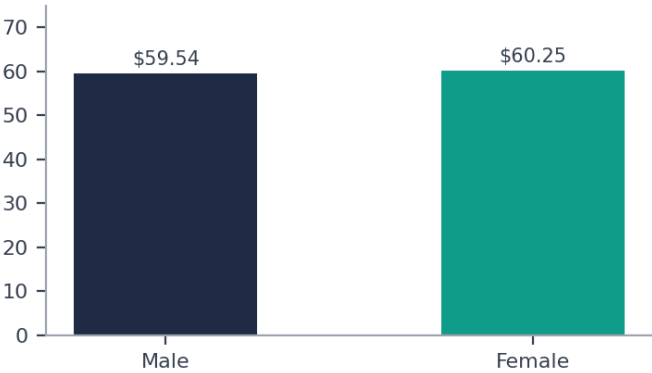
	gender text	total_orders bigint	total_purchase numeric	order_percentage numeric
1	Male	2652	157890	68.00
2	Female	1248	75191	32.00

Screenshot 1: gender-wise orders, total purchase and order percentage

Revenue share by gender



Average purchase amount



Insight

Men are more than twice as many as women, but women spend about \$0.71 more per purchase. The gap in revenue comes from customer volume, not from spending power, so the female segment is an under-served opportunity.

1.2 Age group

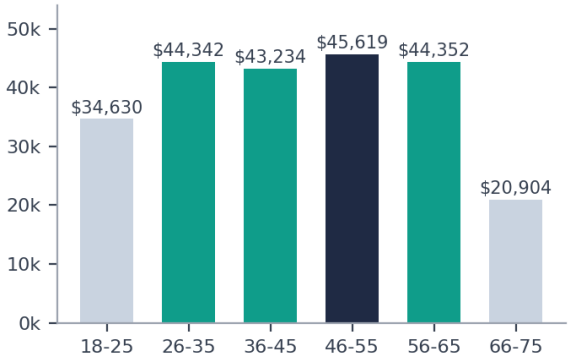
	age_group text	total_customers bigint	avg_purchase_amount numeric	total_revenue numeric
1	18-25	571	60.65	34630
2	26-35	742	59.76	44342
3	36-45	729	59.31	43234
4	46-55	753	60.58	45619
5	56-65	750	59.14	44352
6	66-75	355	58.88	20904

Screenshot 8: age group - customers, avg purchase and revenue

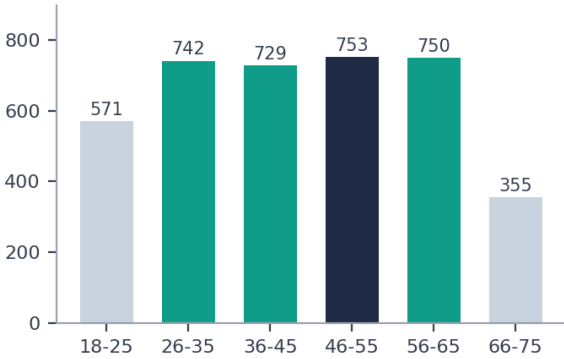
	age_group text	total_contribution numeric
1	36-45	43234
2	66-75	20904
3	46-55	45619
4	26-35	44342
5	18-25	34630
6	56-65	44352

Screenshot 13: revenue contribution by age group

Revenue by age group



Customers by age group



Insight

Customers aged 26 to 65 are evenly split and together bring 76.2% of revenue. The 46-55 group is the top contributor. The 18-25 group has the **highest average purchase (\$60.65)** but a smaller base, so it is a good target for growth. The 66-75 group is only half the size of the others and has the lowest average spend.

2. Loyalty, Subscription and Purchase Frequency

2.1 Customer segments and subscription

	customer_segment text	total_customer bigint
1	Loyal	3116
2	New	83
3	Returning	701

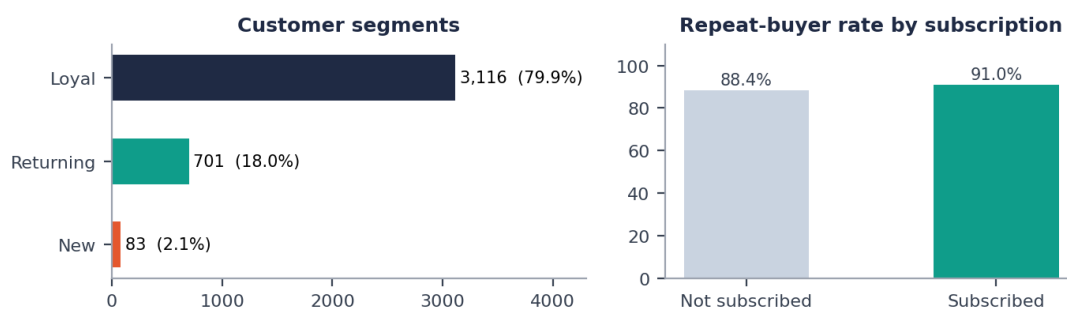
Screenshot 11: customer segments

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Screenshot 12: repeat buyers by subscription

	subscription_status text	total_orders bigint	total_amount numeric	avg_purchase_amount numeric
1	No	2847	170436	59.87
2	Yes	1053	62645	59.49

Screenshot 3: subscription status - orders and purchase amount



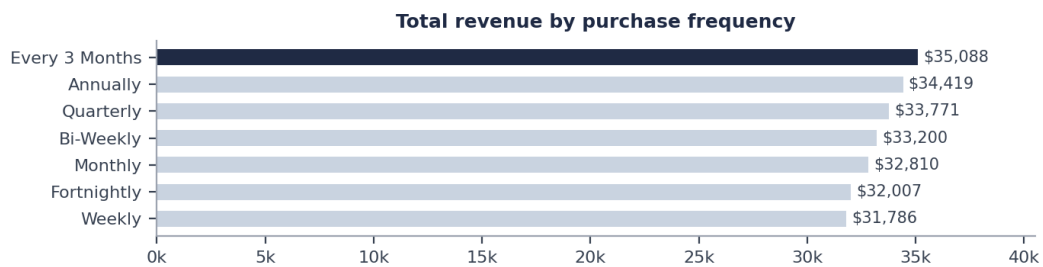
Insight

Nearly four out of five customers are Loyal, which is a major strength. But with only 2.1% new customers, the business is not refreshing its base. Subscribers repeat more often (91.0% vs 88.4%) yet spend the same per purchase, and 73% of customers are still not subscribed. Subscription perks that encourage **bigger baskets** could add value.

2.2 Purchase frequency

	purchase_frequency text	total_customers bigint	avg_previous_purchases numeric	avg_purchase_amount numeric	total_revenue numeric
1	Quarterly	563	26.85	59.98	33771.00
2	Weekly	539	25.77	58.97	31786.00
3	Monthly	553	25.28	59.33	32810.00
4	Fortnightly	542	25.27	59.05	32007.00
5	Every 3 Months	584	24.96	60.08	35088.00
6	Bi-Weekly	547	24.79	60.69	33200.00
7	Annually	572	24.56	60.17	34419.00

Screenshot 6: purchase frequency - customers, purchases and revenue



Insight

Customers are spread almost equally over all seven frequencies (539 to 584 each), and revenue ranges only from 13.6% to 15.1%. Weekly and fortnightly shoppers do **not** spend more than occasional ones. Bi-Weekly buyers have the highest average purchase (\$60.69), and Quarterly buyers have the most previous purchases (26.85).

3. Product and Seasonal Analysis

3.1 Top items by number of orders and by revenue

	category text	item purchased text	total_orders bigint	rank bigint
1	Accessori...	Jewelry	171	1
2	Accessori...	Sunglasses	161	2
3	Accessori...	Belt	161	2
4	Accessori...	Scarf	157	3
5	Clothing	Blouse	171	1
6	Clothing	Pants	171	1
7	Clothing	Shirt	169	2
8	Clothing	Dress	166	3
9	Footwear	Sandals	160	1
10	Footwear	Shoes	150	2
11	Footwear	Sneakers	145	3
12	Outerwear	Jacket	163	1
13	Outerwear	Coat	161	2

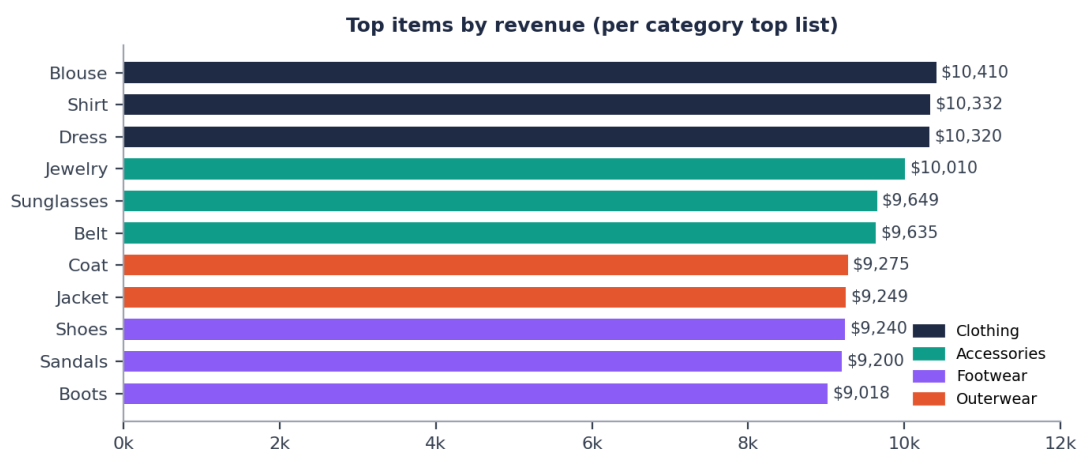
Screenshot 2: top items by number of orders

Note: the Outerwear rank 3 item was not visible in the source screenshots.

	category text	item purchased text	total_orders numeric	rank bigint
1	Accessori...	Jewelry	10010	1
2	Accessori...	Sunglasses	9649	2
3	Accessori...	Belt	9635	3
4	Clothing	Blouse	10410	1
5	Clothing	Shirt	10332	2
6	Clothing	Dress	10320	3
7	Footwear	Shoes	9240	1
8	Footwear	Sandals	9200	2
9	Footwear	Boots	9018	3
10	Outerwear	Coat	9275	1
11	Outerwear	Jacket	9249	2

Screenshot 4: top items by revenue (column labelled total_orders)

3.2 Revenue by top item



Insight

Jewelry and **Blouse** rank first in their category on both orders and revenue. Blouse, Shirt and Dress are the three biggest Clothing earners (all above \$10,300). In Footwear and Outerwear the leader changes depending on the metric (Sandals leads orders but Shoes leads revenue; Jacket leads orders but Coat leads revenue), which means the items in these categories perform very closely. Among items in both lists, revenue per order is highest for Dress (\$62.17) and lowest for Jacket (\$56.74).

3.3 Top items by season

	season text	item purchased text	totalOrders bigint	rank bigint
1	Fall	Jacket	54	1
2	Fall	Hat	50	2
3	Fall	Handbag	48	3
4	Spring	Sweater	52	1
5	Spring	Shorts	47	2
6	Spring	Skirt	46	3
7	Spring	Blouse	46	3
8	Spring	Coat	46	3
9	Summer	Pants	50	1
10	Summer	Dress	47	2
11	Summer	Jewelry	47	2
12	Summer	Shoes	46	3
13	Winter	Sunglasses	52	1
14	Winter	Pants	51	2
15	Winter	Shirt	50	3

Screenshot 5: top items per season

Insight

Fall is led by Jacket and Spring by Sweater, which fits the weather. **Pants** appear in both Summer and Winter, making them a year-round staple. Winter's top item being Sunglasses is unusual and is worth investigating. Counts differ by only 1 to 8 orders between items, so seasonal preference is weak and demand is spread widely across the catalogue.

4. Discounts and Promotions

	discount applied text	promo code used text	total_customers bigint	avg_purchase_amount numeric	total_revenue numeric
1	Yes	Yes	1677	59.28	99411
2	No	No	2223	60.13	133670

Screenshot 7: discount and promo code usage



Insight

Discount and promo code always occur together (both flags match on every customer). Customers using them spend \$0.85 (1.4%) **less** per purchase, and more than half of revenue comes from full-price buyers. This suggests discounts are given to people who would already buy, which reduces margin without raising the basket. Targeted, conditional offers (for example a minimum spend) would be more effective.

5. Payment Method and Shipping Type

5.1 Payment method

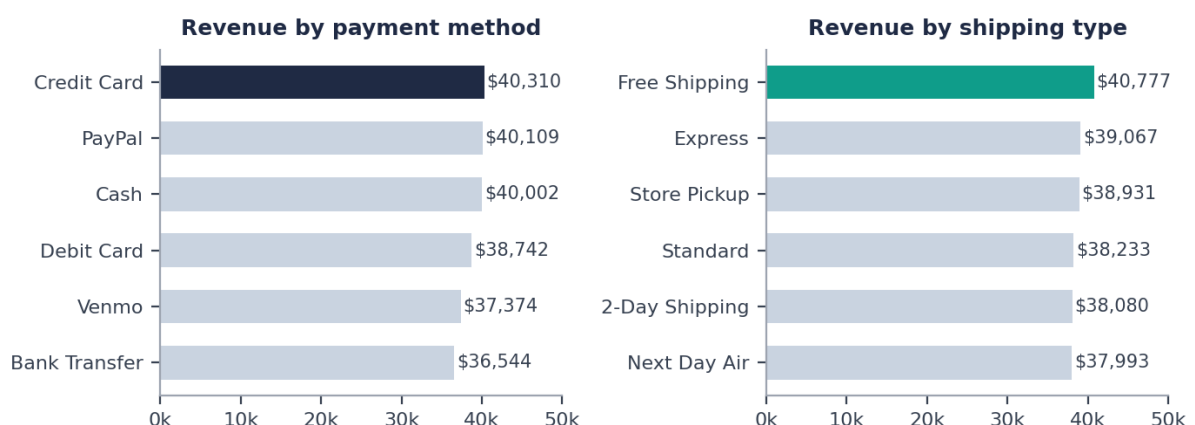
	payment method text	total_customers bigint	sum numeric
1	PayPal	677	40109
2	Credit Card	671	40310
3	Cash	670	40002
4	Debit Card	636	38742
5	Venmo	634	37374
6	Bank Transfer	612	36544

Screenshot 9: payment method - customers and revenue

5.2 Shipping type

	shipping type text	total_customers bigint	avg_purchase_amount numeric	total_revenue numeric
1	2-Day Shipping	627	60.73	38080.00
2	Express	646	60.48	39067.00
3	Free Shipping	675	60.41	40777.00
4	Store Pickup	650	59.89	38931.00
5	Next Day Air	648	58.63	37993.00
6	Standard	654	58.46	38233.00

Screenshot 10: shipping type - customers, avg purchase and revenue



Insight

Payment methods are almost equally used, with each one between 15.7% and 17.4% of customers. Credit Card brings the most revenue (\$40,310) and Bank Transfer the least (\$36,544). For shipping, **Free Shipping** is the most popular and earns the most. 2-Day (\$60.73) and Express (\$60.48) customers spend the most per purchase, while Standard (\$58.46) and Next Day Air (\$58.63) customers spend the least.

6. Recommendations

1. Invest in new-customer acquisition

Only 2.1% of customers are New. Launch acquisition campaigns and referral rewards so the loyal base is not the only source of growth.

2. Grow the female customer segment

Women are 32% of customers but spend slightly more per purchase. Targeted campaigns and product ranges could lift revenue with little extra cost.

3. Rethink blanket discounts

Discount users spend less, and 57.3% of revenue is already full-price. Replace broad discounts with conditional offers such as minimum-spend or bundle deals.

4. Make subscription more valuable

73% of customers are not subscribed. Subscribers repeat more but do not spend more, so add perks that increase basket size (free shipping, early access, bundles).

5. Use free shipping strategically

Free Shipping is the most popular option and the top earner. Test a free-shipping threshold slightly above the \$60 average purchase to raise basket value.

6. Target the edges of the age range

18-25 customers spend the most per purchase but are a smaller group, and 66-75 is the smallest group. Tailored marketing for both could widen the customer base.

7. Keep stock depth on proven items

Prioritise availability of Jewelry, Blouse, Pants, Shirt, Dress, Jacket, Sandals and Shoes, and plan seasonal stock (Jackets in Fall, Sweaters in Spring, Pants year-round).

8. Keep all payment options

Usage is balanced across all six methods, so there is no clear method to remove. Bank Transfer and Venmo are the smallest and could be promoted at checkout.

Data Notes and Limitations

- The dataset has 3,900 records, and every table adds up to 3,900 customers and \$233,081 in revenue. Each record is treated as one customer and one purchase. Amounts are assumed to be in USD.
- The source table for top items by revenue labels its value column *total_orders*, but the values (about \$9,000 to \$10,400 per item) match revenue, so they are reported here as revenue.
- Some ranked tables were cropped in the screenshots (for example Outerwear rank 3), so only visible rows are shown.
- Differences between many groups are small (often under 3%). Treat them as directional signals and confirm with statistical tests before making large business decisions.
- The repeat-buyer count comes from the source query; its exact definition was not shown.